

EDA-Data Import and Export

Code:

```
import pandas as pd
import sqlite3

# Read CSV
df = pd.read_csv("StudentsPerformance.csv")

# Add data using Series
s = pd.Series({
    "gender": "female",
    "race/ethnicity": "group B",
    "parental level of education": "bachelor's degree",
    "lunch": "standard",
    "test preparation course": "completed",
    "math score": 85,
    "reading score": 90,
    "writing score": 88
})

# Add data using DataFrame
new_df = pd.DataFrame([
    ["male", "group C", "master's degree", "standard", "none", 78, 80, 76],
    ["female", "group D", "associate's degree", "free/reduced", "completed", 92, 95, 93]
], columns=df.columns)

# Append Series and DataFrame
df = pd.concat([df, pd.DataFrame([s]), new_df], ignore_index=True)

# Export to CSV and Excel
df.to_csv("StudentsPerformance_updated.csv", index=False)
df.to_excel("StudentsPerformance_updated.xlsx", index=False)

# Store in SQL
conn = sqlite3.connect("students.db")
df.to_sql("StudentsPerformance", conn, if_exists="replace", index=False)
```

```
# Show last 5 rows from SQL table
```

```
result = pd.read_sql(  
    "SELECT * FROM StudentsPerformance LIMIT 10",  
    conn  
)
```

```
print(result)  
conn.close()
```

Output:

The screenshot displays a JupyterLab environment with a file explorer on the left, a central code editor, and an Anaconda Assistant sidebar on the right. The code editor shows a Python script that reads a CSV file, prints the first 5 rows, and prints the last 5 rows. The output of the script is displayed in the console, showing the first 5 rows and the last 5 rows of the dataset.

```
[2]: import pandas as pd  
df = pd.read_csv("StudentsPerformance_updated.csv")  
print(df.head()) # First 5 rows  
print(df.tail()) # Last 5 rows
```

	gender	race/ethnicity	parental level of education	lunch	
0	female	group B	bachelor's degree	standard	
1	female	group C	some college	standard	
2	female	group B	master's degree	standard	
3	male	group A	associate's degree	free/reduced	
4	male	group C	some college	standard	

```
test preparation course math score reading score writing score  
0 none 72 72 74  
1 completed 69 90 88  
2 none 90 95 93  
3 none 87 57 44  
4 none 76 78 75
```

	gender	race/ethnicity	parental level of education	lunch	
998	female	group D	some college	standard	
999	female	group D	some college	free/reduced	
1000	female	group B	bachelor's degree	standard	
1001	male	group C	master's degree	standard	
1002	female	group D	associate's degree	free/reduced	

```
test preparation course math score reading score writing score  
998 completed 68 78 77  
999 none 77 86 86  
1000 completed 85 90 88  
1001 none 78 80 76  
1002 completed 92 95 93
```

```
[4]: import pandas as pd  
df = pd.read_excel("StudentsPerformance_updated.xlsx")  
print(df.head())  
print(df.tail())
```

	gender	race/ethnicity	parental level of education	lunch	
0	female	group B	bachelor's degree	standard	
1	female	group C	some college	standard	
2	female	group B	master's degree	standard	
3	male	group A	associate's degree	free/reduced	
4	male	group C	some college	standard	

localhost:8888/notebooks/Untitled4.ipynb?

Tools for easy setup and effortless collaboration.

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Manage Code Snippets

Environments

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AI Assistant

Chat-based Python Help

Jupyter Untitled4 Last checkpoint: 37 minutes ago

File Edit View Run Kernel Settings Help

Python 3 (ipykernel)

```
# Create SQLite database
conn = sqlite3.connect("students.db")

# Store DataFrame in SQL table
df.to_sql("StudentsPerformance", conn, if_exists="replace", index=False)

# Display first 10 rows from SQL table
sql_table = pd.read_sql("""
SELECT *
FROM StudentsPerformance
LIMIT 10
""", conn)

display(sql_table)

conn.close()
```

	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score
0	female	group B	bachelor's degree	standard	none	72	72	74
1	female	group C	some college	standard	completed	69	90	88
2	female	group B	master's degree	standard	none	90	95	93
3	male	group A	associate's degree	free/reduced	none	47	57	44
4	male	group C	some college	standard	none	76	78	75
5	female	group B	associate's degree	standard	none	71	83	78
6	female	group B	some college	standard	completed	88	95	92
7	male	group B	some college	free/reduced	none	40	43	39
8	male	group D	high school	free/reduced	completed	64	64	67
9	female	group B	high school	free/reduced	none	38	60	50

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StudentsPerformance_updated.ipynb

```
gender,race/ethnicity,parental level of education,lunch,test preparation course,math score,reading score,writing score
female,group B,bachelor's degree,standard,none,72,72,74
female,group C,some college,standard,completed,69,90,88
female,group B,master's degree,standard,none,90,95,93
male,group A,associate's degree,free/reduced,none,47,57,44
male,group C,some college,standard,none,76,78,75
female,group B,associate's degree,standard,none,71,83,78
female,group B,some college,standard,completed,88,95,92
male,group B,some college,free/reduced,none,40,43,39
male,group D,high school,free/reduced,completed,64,64,67
female,group B,high school,free/reduced,none,38,60,50
male,group C,associate's degree,standard,none,54,54,52
male,group D,associate's degree,standard,none,40,52,43
female,group B,high school,standard,none,65,81,73
male,group A,some college,standard,completed,78,72,70
female,group A,master's degree,standard,none,50,53,58
female,group C,some high school,standard,none,69,75,78
male,group C,high school,standard,none,88,89,86
female,group B,some high school,free/reduced,none,19,32,28
male,group C,master's degree,free/reduced,completed,46,42,46
female,group C,associate's degree,free/reduced,none,54,58,61
male,group D,high school,standard,none,66,69,63
female,group B,some college,free/reduced,completed,65,75,70
male,group D,some college,standard,none,44,54,53
female,group C,some high school,standard,none,69,73,73
male,group D,bachelor's degree,free/reduced,completed,74,71,80
male,group A,master's degree,free/reduced,none,73,74,72
male,group B,some college,standard,none,69,54,55
female,group C,bachelor's degree,standard,none,67,69,75
male,group C,high school,standard,none,70,70,65
female,group D,master's degree,standard,none,62,70,75
female,group D,some college,standard,none,69,74,74
female,group B,some college,standard,none,63,65,61
female,group E,master's degree,free/reduced,none,56,72,65
male,group D,some college,standard,none,40,43,38
male,group E,some college,standard,none,97,87,82
male,group E,associate's degree,standard,completed,81,81,79
female,group D,associate's degree,standard,none,74,81,83
female,group D,some high school,free/reduced,none,50,64,59
female,group D,associate's degree,free/reduced,completed,75,90,88
male,group B,associate's degree,free/reduced,none,57,56,57
male,group C,associate's degree,free/reduced,none,55,51,54
female,group C,associate's degree,standard,none,58,73,68
female,group B,associate's degree,standard,none,53,58,65
male,group B,some college,free/reduced,completed,59,65,66
female,group E,associate's degree,free/reduced,none,50,56,54
male,group B,associate's degree,standard,none,65,54,57
female,group A,associate's degree,standard,completed,55,65,62
female,group C,high school,standard,none,66,71,76
female,group D,associate's degree,free/reduced,completed,57,74,76
male,group C,high school,standard,completed,82,84,82
```

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